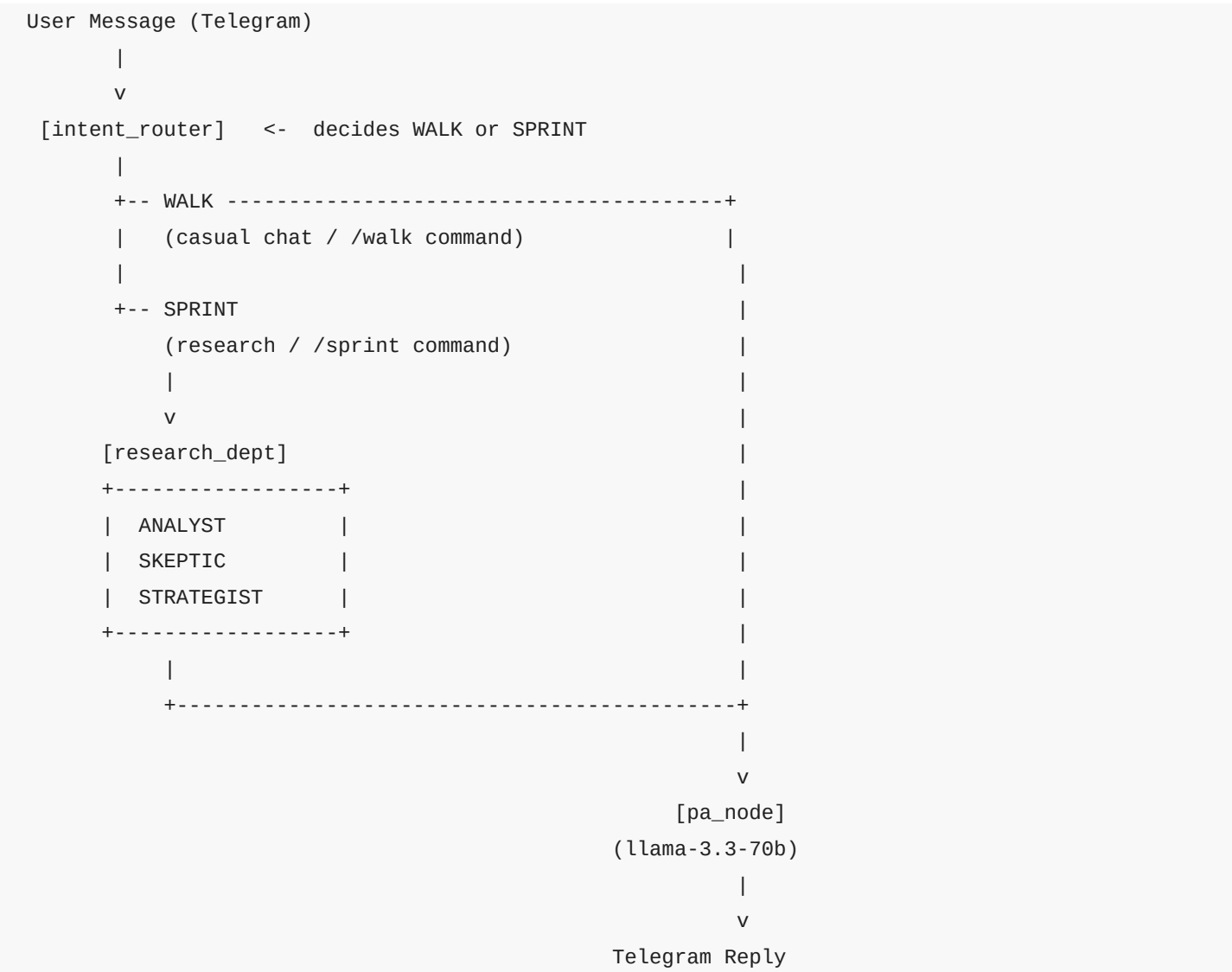


Overview

ARIA is a Telegram bot powered by a LangGraph multi-agent pipeline. It automatically routes incoming messages into two gears - WALK for casual chat and SPRINT for deep research - and uses a 3-agent swarm (Analyst, Skeptic, Strategist) to synthesize thorough, well-rounded responses via a Llama 3 Personal Assistant.

Agent Flow



Gear System

Gear	Trigger	Behaviour
WALK	Casual chat / /walk command	Single PA call - brief and direct
SPRINT	Research query / /sprint cmd	3-agent swarm -> PA synthesis

Tech Stack

Layer	Tool	Free Tier
Bot interface	python-telegram-bot	Free
Orchestration	LangGraph + LangChain	Free
Router + Research	llama-3.1-8b-instant (Groq)	6,000 req/day
PA final reply	llama-3.3-70b-versatile (Groq)	6,000 req/day
Hosting	Replit Reserved VM	Always-on

Required Secrets

Secret	Purpose
GROQ_API_KEY	Groq LLM API key (free at console.groq.com)
TELEGRAM_BOT_TOKEN	Telegram bot token from @BotFather

File Structure

```
aria/  
  bot.py          <- All ARIA logic: router, research dept, PA, Telegram handler  
  generate_pdf.py <- This PDF generator  
  replit.md       <- Project documentation & architecture  
  artifacts/  
    api-server/   <- Deployment config (artifact.toml)
```

Telegram Commands

Input	Behaviour
Any casual message	Auto-routes to WALK (quick reply)
Any research question	Auto-routes to SPRINT (3-agent swarm)
/sprint in message	Forces SPRINT mode regardless of content
/walk in message	Forces WALK mode regardless of content

Important Notes

- 1. Bot uses polling (not webhooks) - must be deployed as a Reserved VM, NOT autoscale.
- 2. Groq free tier: 6,000 requests/day and 30 requests/minute.
- 3. SPRINT mode makes 4 Groq calls per message (1 router + 3 agents + 1 PA).
- 4. LangGraph retry logic may delay error responses by ~30s when rate-limited.